



Apni Chhat Apna Ghar

Field Engineer User Manual

AI-Based Construction Site Validation System

The Urban Unit | Government of Punjab

Version 0.2 | For Field Engineers

Internal guide for assigned project inspections,
progress uploads, AI validation, and construction modules.

1. About ACAG

ACAG (Apni Chhat Apna Ghar) is an AI-based construction site validation app developed with The Urban Unit / Government of Punjab. As a Field Engineer, you manage assigned house-construction projects: inspect sites, verify location, upload progress photos, run AI validation, complete construction modules, and approve or request rework on inspection reports.

2. Getting Started

Sign in

1. Open the app and wait for the splash screen.
2. On Login, choose the Engineer role card.
3. Enter your Email and Password.
4. Tap Login.

Your account role must be Engineer. If you select the wrong role, you will see a message such as: This account is not an Engineer. Select Home Owner.

Sign Up opens a registration form (Full Name, Email, Phone, Password). Use it only if your organization supports self-registration; otherwise use credentials provided by your administrator.

Sign out

Profile -> Logout -> confirm with Logout.

3. Main Navigation

The engineer home screen uses a bottom bar with a center camera button:

Tab	Purpose
Dashboard	Summary, KPIs, and recent alerts
Projects	Your assigned construction sites
Camera (center)	Quick site photo capture
Reports	Past inspection reports
Profile	Account details and settings

4. Dashboard

After login you land on the Dashboard. Typical content includes:

- Welcome message with your name
- KPIs: Assigned, Today's, Pending, Completed
- Inspection Overview / monthly progress
- Recent Notifications (for example: pending inspection, new assignment)

Use Assigned or View All shortcuts to open projects. The notification bell may show a badge; full notification

management may be limited depending on the build.

5. Projects

1. Open the Projects tab.
2. Search by project name, owner, or location.
3. Filter with: All | Pending | In Progress | Completed.
4. Tap a project card to open Project Details.

Projects are already assigned to you. There is no separate Accept Job step.

Project Details hub

- Status badge (for example: PENDING INSPECTION)
- Actions: Contact Owner (opens phone dialer), Start Inspection, View on Map
- Tabs: Details, Images, AI Status, Materials
- Images tab: tap any photo for full-screen preview
- Visual progress stages: DPC -> Brick Work -> Roof -> Plaster -> Finishing
- Five construction modules (Module 4 detailed in Section 9)
- Add photos from the app bar camera icon or Add Photo

Project data (modules, images, materials, owner phone) loads from Supabase when you open a project. View on Map opens the project location in an external maps application.

6. Core Inspection Workflow

This is the main field job loop for site validation:

1. Open the project -> tap Start Inspection.
2. GPS Verification - wait until status shows verified (for example: Verified within 12m), then continue.
3. Upload Progress - select stage, add notes, attach a photo (Take Photo or Choose from Gallery), then tap Submit Progress.

Construction stage options:

- Site Clearance
 - Foundation
 - Plinth Beam
 - Brick Work
 - Roof Casting
 - Finishing
4. AI Validation - review score and checklist items (GPS match, stage visual match, image quality, duplicate detection, safety hazards). Result may show as Pass with Warning.
 5. Tap Generate Report.
 6. On Inspection Report, choose Approve, Request Rework, or Done.

Action	Meaning
Approve	Accept the inspection outcome
Request Rework	Send the finding back for corrections
Done	Return to the main engineer screens

You can reopen reports later from the Reports tab. Share and download options are available on the report screen.

7. Quick Photo Capture

1. Tap the center camera button on the bottom bar.
2. Choose Take Photo or Choose from Gallery.
3. The photo is saved to your primary/current project as a quick capture.

Allow camera and photo library access when prompted. These permissions are required for site photos and profile pictures.

8. Construction Modules (Overview)

From Project Details, open modules in sequence. Each completed module advances overall project progress by approximately 20% (5 modules total).

Module	What you complete
01 Planning & Elevation	Plot dimensions, rooms, floor plans, elevation
02 Foundation & Structural	Stories, soil analysis, foundation drawing, frame type
03 Material Estimation	Material quantities and cost breakdown
04 Construction Tracking	Stage timeline, progress photos, quality review (see Section 9)
05 Handover & HSE	Handover, HSE compliance, completion certificate

Tap Start Module on a module hub screen, or open Module 4 directly from Project Details. Progress and module checkmarks refresh from the database when you return to Project Details.

9. Module 4 - Construction Tracking & QA (Detailed)

Module 4 is the main on-site progress workflow. It tracks seven construction stages, saves photos and descriptions to the database, and shows completed work on the Project Details screen.

9.1 Seven construction stages

Stages unlock one at a time in this order:

1. Foundation & Plinth
2. Structure (Columns & Beams)
3. Brickwork & Plaster
4. Roof Slab & Waterproofing
5. Electrical & Plumbing
6. Finishing & Paint
7. Final Inspection & Handover

9.2 Open Module 4

1. Open Project Details for your assigned project.
2. Scroll to Construction Modules and tap "04 Construction Tracking".

3. Tap Start Module to open the Stage Timeline screen.

On first open, no stages are checked. The progress bar shows 0 of 7 stages complete.

9.3 Complete one stage (full flow)

1. On Stage Timeline, tap "Upload Progress Photos".
2. Photo Upload - take a photo (camera) or choose from gallery. Add an optional description, then tap Next - Quality Assessment.
3. Quality Assessment - review the QA score and checklist. Tap Complete.
4. You return to Project Details. Stage 1 is now marked DONE on the timeline.
5. Open the Images tab on Project Details - your uploaded photo appears there. Tap any image for a full-screen preview with pinch-to-zoom.

Repeat the upload flow for each new visit. Each completion checks the next stage (Stage 2, then Stage 3, and so on). After all 7 stages are done, Module 4 is marked complete and the upload button is disabled.

9.4 View completed stage details

1. Re-open Module 4 -> Stage Timeline.
2. Tap any stage marked DONE (green check).
3. The stage detail screen shows the saved photo and description.
4. Tap the image for preview, or use Back to return.

9.5 Contact Owner from Project Details

On Project Details, tap Contact Owner to open the phone dialer with the owner's number (from the project record or owner profile). Ensure a valid Pakistan-format number is stored (for example +923001234567).

9.6 Module 4 troubleshooting

Issue	What to do
Table / save error	Ask admin to confirm module04_construction_stages exists in Supabase.
Stages not updating	Pull to refresh on Project Details, or re-open the project.
No images in Images tab	Complete the Quality Assessment step; photos save on Complete.
Contact Owner does nothing	Set owner_phone on the project or phone on the owner profile.

10. Other Modules (Brief)

- Module 01: Plot size, rooms, floor plans, elevation design.
- Module 02: Stories, soil analysis, foundation, structural frame.
- Module 03: Material quantities and PKR cost breakdown.
- Module 05: Handover checklist, HSE compliance, certificate.

11. Reports

The Reports tab lists past inspections. Open a report to review findings and scores, and use share/download when you need an offline copy.

12. Profile

- View name, contact number, location, and Field Engineer badge
- Change profile photo using the camera icon on the avatar
- Edit Profile (Settings): update Full Name, Contact Number, and Location
- Other settings: Notifications, Change Password, Help & Support, About ACAG
- Logout to end your session

Profile photo, name, phone, and location are saved to your Supabase profile. Some settings rows may be placeholders depending on the build.

13. Demo / Test Data

Use these seeded accounts to test the app end-to-end. Password for both: 12345678

Field	Engineer	Home Owner
Name	Shoaib Khilji	Ali Raza
Email	shoaibkhilji141@gmail.com	ali.raza.owner@gmail.com
Location	Lahore, Punjab	Raiwind Road, Lahore
Contact	Set in Profile (Edit Profile)	Set in Profile (Edit Profile)
Profile photo	Camera icon on Profile	Camera icon on Profile

Test project (ACAG-1)

Field	Value
Project code / name	ACAG-1 / House #ACAG-1
Project location	Johar Town, Block A, Lahore
Owner name	Ali Raza
Owner contact	projects.owner_phone or owner profile phone
Assigned engineer	Shoaib Khilji

Before testing Module 4 Contact Owner, add phone numbers via Edit Profile (engineer & owner) and/or set owner_phone on project ACAG-1 in Supabase. Use format +923XXXXXXXXX for dialer testing.

14. Permissions Checklist

Permission	Why it is needed
Camera	Site photos and profile picture
Photos / gallery	Choose existing images to upload
Internet	Sign-in and sync projects / uploads
Maps (external)	View on Map for project location

15. Typical Day - Quick Reference

1. Log in as Engineer and check the Dashboard.
2. Open Projects and select today's site (ACAG-1).
3. Start Inspection -> GPS -> upload stage + photo -> AI -> report -> Approve or Request Rework.
4. Open Module 4: upload progress photo -> Quality Assessment -> Complete. Verify stage checked and image in Images tab.
5. Complete any other due construction modules.
6. Use the center camera for ad-hoc site shots.
7. Update Profile contact/location if needed; test Contact Owner.
8. Logout from Profile when finished.

End of manual. For support, contact your ACAG / Urban Unit project administrator.